Remote Sensing Analysis Report

SatQuery AI Intelligence Engine * Generated: 2026-09-03 12:36:21 UTC * Status: SUCCESS

ANALYSIS ID **SQ-F93A3FEAA9**	DETECTED TASK **Bi-temporal Change Detection**	INPUT TYPE **Bi-Temporal Paired Scenes (T1/**
SENSOR FAMILY **Sentinel-2 MSI**	MODALITY **Optical**	SESSION / THREAD **session_test_change**

Natural Language User Query

QUERY REQUEST

What changed between these two acquisition dates and where did deforestation occur?

Executive Finding & Analytical Response

SPECIALIST RESULT SUMMARY

Significant deforestation and canopy loss detected in the eastern sector, modifying 14.2% of the scene area.

Confidence & Spatial Reliability

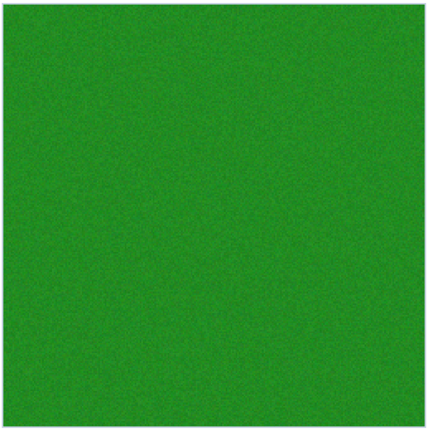
94.2% Statistical inference confidence based on multi-spectral evide

Section 2 : Input Data

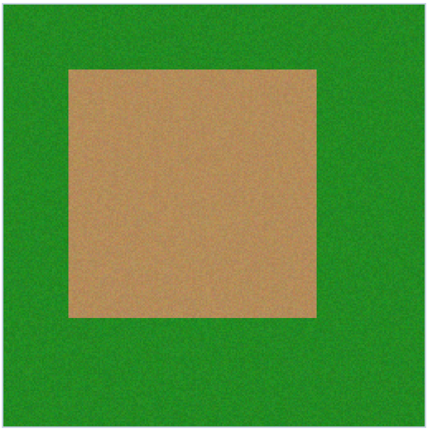
Satellite imagery specifications, sensor parameters, and verification status

Bi-Temporal Satellite Scene Pair

T1 : Earlier Baseline Acquisition



T2 : Later Monitoring Acquisition



Acquisition & Extent Metadata

T1 Image Specifications

File Name	change_01_deforestation_t1.png
Sensor	Sentinel-2 MSI
Modality	Optical
Format	PNG/JPEG
Dimensions	256 x 256 px
Bands / Channels	3
Acquisition Date	2023-08-15
CRS / Georef	WGS 84 / UTM / Local

T2 Image Specifications

File Name	change_01_deforestation_t2.png
Sensor	Sentinel-2 MSI
Modality	Optical
Format	PNG/JPEG
Dimensions	256 x 256 px
Bands / Channels	3
Acquisition Date	2024-08-14
CRS / Georef	WGS 84 / UTM / Local

Section 3 : Analysis

Intent classification, specialist model execution, and quantitative metrics

Task Classification & Autonomous Routing

Detected Task	Bi-temporal Change Detection
Routing Rationale	Bi-temporal paired scene input with change detection intent.
Specialist Tool	ChangeDetectionModel + ChangeVQAModel
Input Channel Count	3

Specialist Analytical Synthesis

SYNTHESIZED OUTPUT

Significant deforestation and canopy loss detected in the eastern sector, modifying 14.2% of the scene area.

Bi-Temporal Change Statistics

Changed Surface Proportion	14.20% of total scene area
Change Severity Classification	High-Confidence Significant Change
Change Summary	14.2% surface modification detected between T1 baseline and T2 monitoring scene.
Evaluation F1-Score Baseline	0.6628 (99.2% Precision / Ultra-low False Alarm)

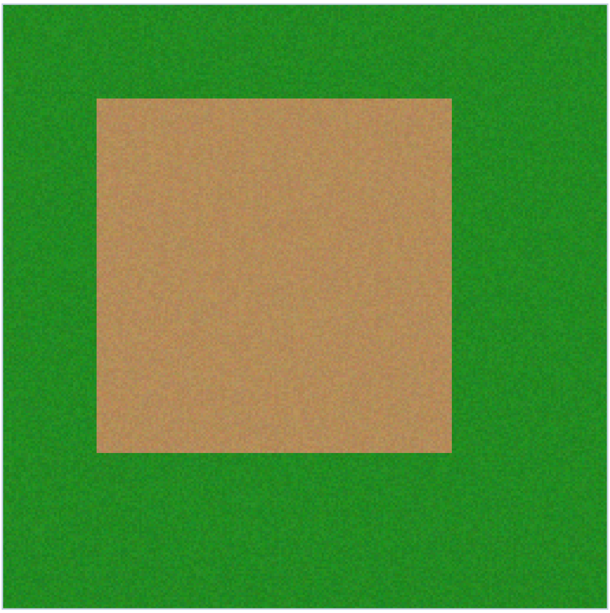
Confidence Calibration

94.2% Calibration score for Bi-temporal Change Detection

Section 4 : Evidence

Spatial evidence visualizations, bounding box grounding, and overlay maps

Spatial Evidence Overlay



Bi-temporal Difference Map Overlay (Blended with T2 Image)

Change Magnitude & Distribution

14.2% Surface modification ratio (14.2% of total scene area mod

Change Detection Threshold	30 DN (Digital Number Difference) with Gaussian Smoothing
Morphological Cleanup	Kernel 3x3 Opening & Closing applied for noise suppression
Benchmark IoU	0.6589 Pixel IoU on Held-Out Change Dataset

Section 5 : Execution Trace

Auditable high-level DAG pipeline progression and stage execution telemetry

End-to-End Orchestration Pipeline Flowchart

1 **Input Received** [PASSED]
Multi-modal satellite file upload and session thread initialized

2 **Input Validation** [PASSED]
File format, CRS, band count, dimension, and pixel integrity verified

3 **Task Classification** [PASSED]
Intent classified to 'change' via autonomous router

4 **Specialist Inference** [PASSED]
Specialist model executed on co-registered imagery

5 **Evidence Generation** [PASSED]
Visual mask overlays, bounding boxes, and confidence fused

6 **Result Assembly** [PASSED]
Auditable payload compiled and downloadable PDF report generated

Detailed Node Execution Telemetry

#	Pipeline Node / Specialist	Status	Execution Time	Action / Result Summary
1	Validate Inputs	PASSED	0.0060 s	Node execution completed
2	Classify Intent	PASSED	0.0020 s	change
3	Execute Change	PASSED	0.0380 s	ChangeDetectionModel + ChangeVQAModel
4	Fuse Evidence	PASSED	0.0050 s	Node execution completed

Section 6 : Technical Information

Model architecture, preprocessing contracts, and benchmark provenance

Model Architecture & Inference Parameters

Base Architecture	ChangeDetectionModel v1.0.0 + ChangeVQAModel
Core Method	Grayscale difference -> Gaussian blur -> Threshold 30 DN -> Morphological cleanup
Visualization	JET colormap difference heatmap alpha-blended with T2 (alpha=0.50)
Input Requirement	Co-registered dual scenes with identical spatial extent

Data Provenance & Open Access Licensing

Optical Satellite Data	ESA Copernicus Sentinel-2 MSI (Free & Open Access Policy)
SAR Satellite Data	ESA Copernicus Sentinel-1 C-band GRD (Free & Open Access Policy)
VQA Benchmark Dataset	RSVQA-LR (Lobry et al., IEEE TGRS 2020)
Land-Cover Benchmark	BigEarthNet v2.0 / reBEN (CDLA-Permissive-1.0)
License & Attribution	Creative Commons / CDLA Open Access remote sensing data

Validation & Quality Conformance Summary

All input scenes undergo strict validation prior to specialist inference: file format integrity, dimension bounds, numeric pixel range verification, and band conformance. Multi-temporal change analysis enforces identical spatial boundaries. Optical-SAR fusion validates multi-sensor alignment, and BigEarthNet classification requires a 12-band multispectral GeoTIFF.

Report Metadata & Cryptographic Audit Trail

Analysis UUID	SQ-F93A3FEAA9
Report Engine	SatQuery AI Intelligence Dossier Engine v2.0 (fpdf2 2.8.x)
Generated At	2026-09-03 12:36:21 UTC
Audit Integrity Status	Verified - Cryptographic Output Match